

Expt 4	Design of Low Pass Filter using Windowing Techniques
Date:	4.1. Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique in MATLAB

Code:

```

clc;
clear;
close all;

%-----
% EX NO : 4
% TITLE : Design and Analysis of FIR Low-Pass Filter
%         Using Different Window Techniques
%-----

% Filter Specifications
fp = 1000;    % Passband Cutoff Frequency (Hz)
Fs = 4000;    % Sampling Frequency (Hz)
N = 50;      % Filter Order (Even Number Preferred)

% Normalized Cutoff Frequency
Wc = fp / (Fs/2);

%-----
% FIR Low-Pass Filter Using Rectangular Window
%-----
b_rect = fir1(N, Wc, 'low', rectwin(N+1));

%-----
% FIR Low-Pass Filter Using Hamming Window
%-----
b_hamm = fir1(N, Wc, 'low', hamming(N+1));

%-----
% FIR Low-Pass Filter Using Hanning Window
%-----
b_hann = fir1(N, Wc, 'low', hann(N+1));

%-----
% Display Filter Coefficients
%-----
fprintf('-----\n');
fprintf('Rectangular Window Coefficients\n');
fprintf('-----\n');
disp(b_rect);

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fprintf('-----\n');
fprintf('Hamming Window Coefficients\n');
fprintf('-----\n');
disp(b_hamm);

fprintf('-----\n');
fprintf('Hanning Window Coefficients\n');
fprintf('-----\n');
disp(b_hann);

%-----
% Frequency Response - Rectangular Window
%-----
figure;
freqz(b_rect, 1, 1024, Fs);
title('FIR Low-Pass Filter Using Rectangular Window');

%-----
% Frequency Response - Hamming Window
%-----
figure;
freqz(b_hamm, 1, 1024, Fs);
title('FIR Low-Pass Filter Using Hamming Window');

%-----
% Frequency Response - Hanning Window
%-----
figure;
freqz(b_hann, 1, 1024, Fs);
title('FIR Low-Pass Filter Using Hanning Window');

%-----
% Impulse Response Comparison
%-----
figure;

subplot(3,1,1);
impz(b_rect,1);
title('Impulse Response - Rectangular Window');

subplot(3,1,2);
impz(b_hamm,1);
title('Impulse Response - Hamming Window');

subplot(3,1,3);
impz(b_hann,1);
title('Impulse Response - Hanning Window');

%-----

```

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% Pole-Zero Plots
%-----

figure;

subplot(1,3,1);
zplane(b_rect,1);
title('Rectangular Window');

subplot(1,3,2);
zplane(b_hamm,1);
title('Hamming Window');

subplot(1,3,3);
zplane(b_hann,1);
title('Hanning Window');

%-----
% Display Filter Details
%-----
fprintf('-----\n');
fprintf('FIR Low-Pass Filter Specifications\n');
fprintf('-----\n');
fprintf('Filter Order          = %d\n', N);
fprintf('Sampling Frequency (Fs) = %d Hz\n', Fs);
fprintf('Cutoff Frequency (Fp)   = %d Hz\n', fp);
fprintf('Normalized Cutoff (Wc)  = %.4f\n', Wc);
fprintf('-----\n');

```

Output:

Fig 1:

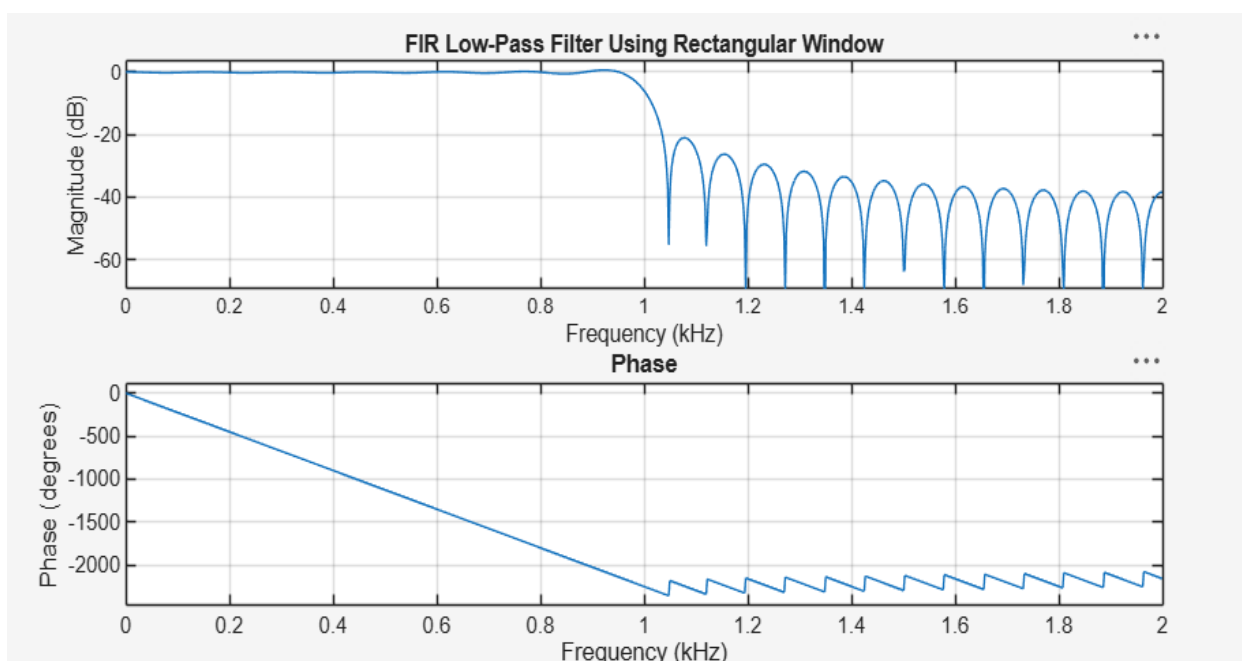


Fig 2:

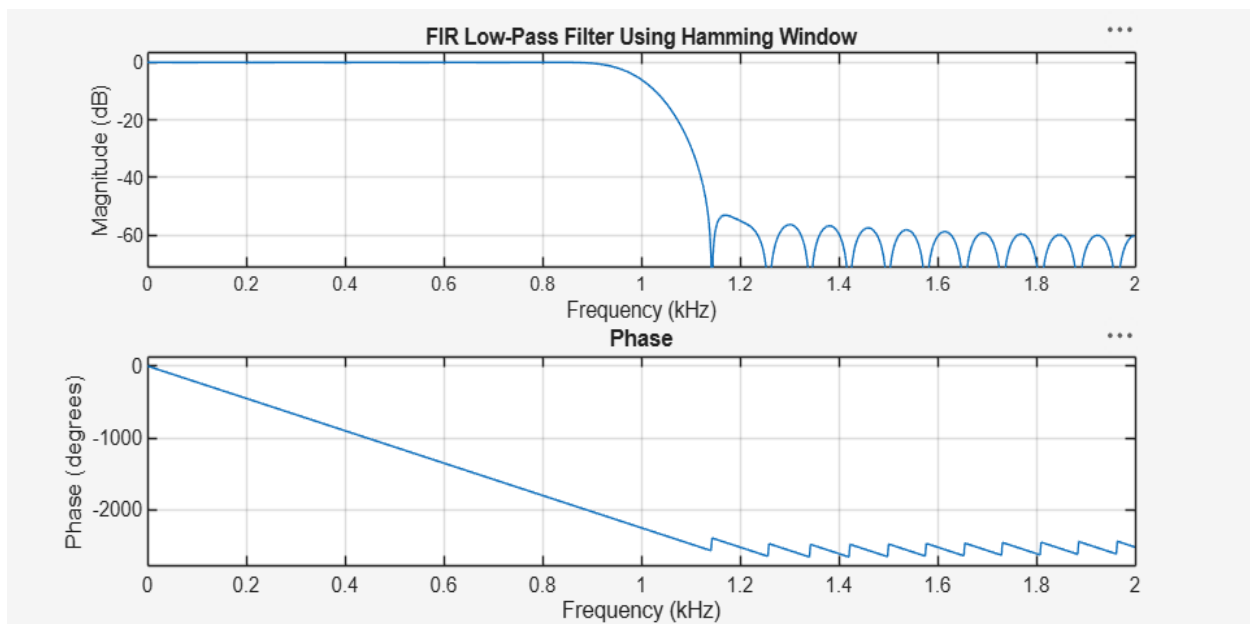


Fig 3:

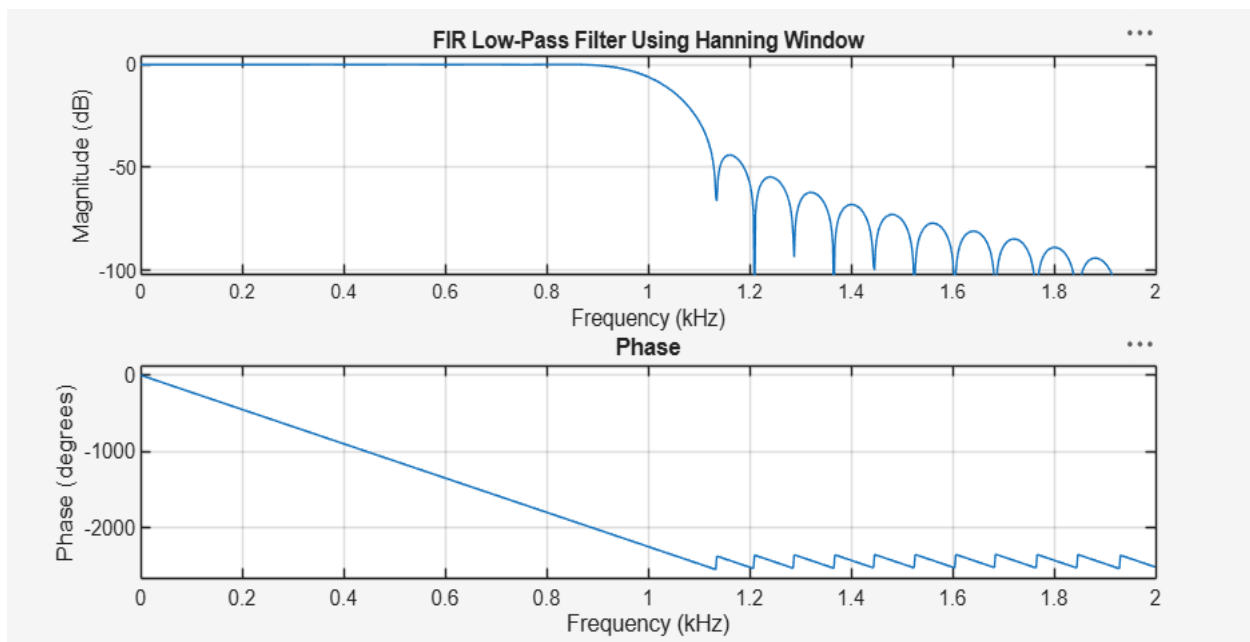


Fig 4:

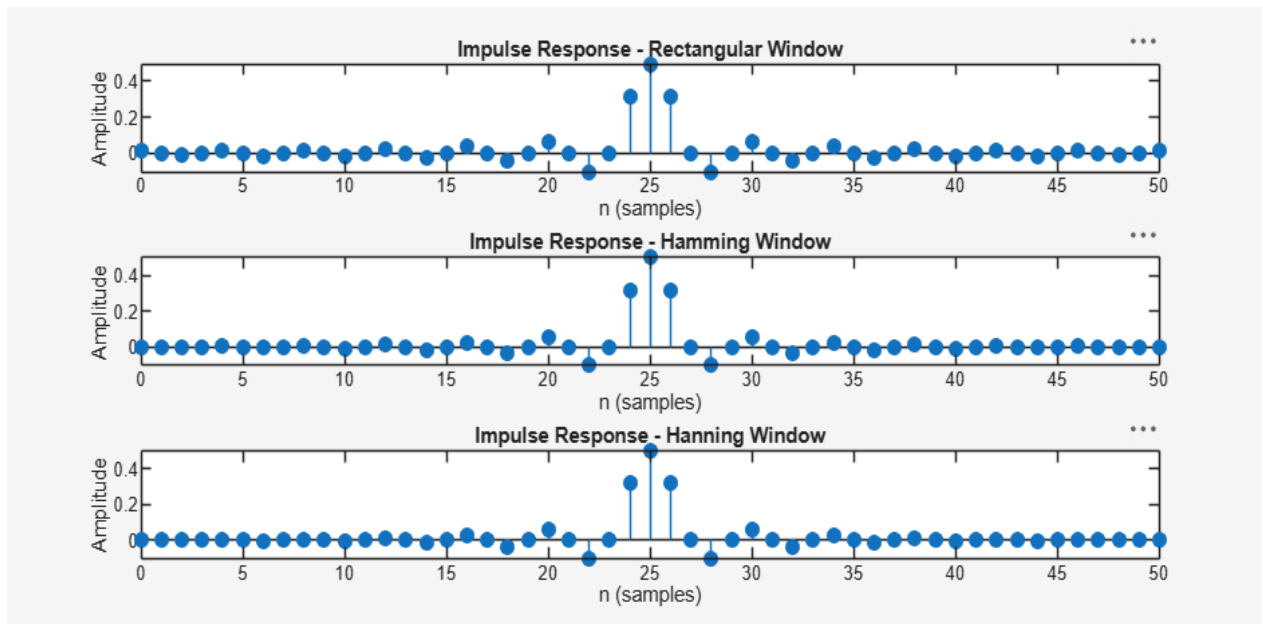


Fig 5:

